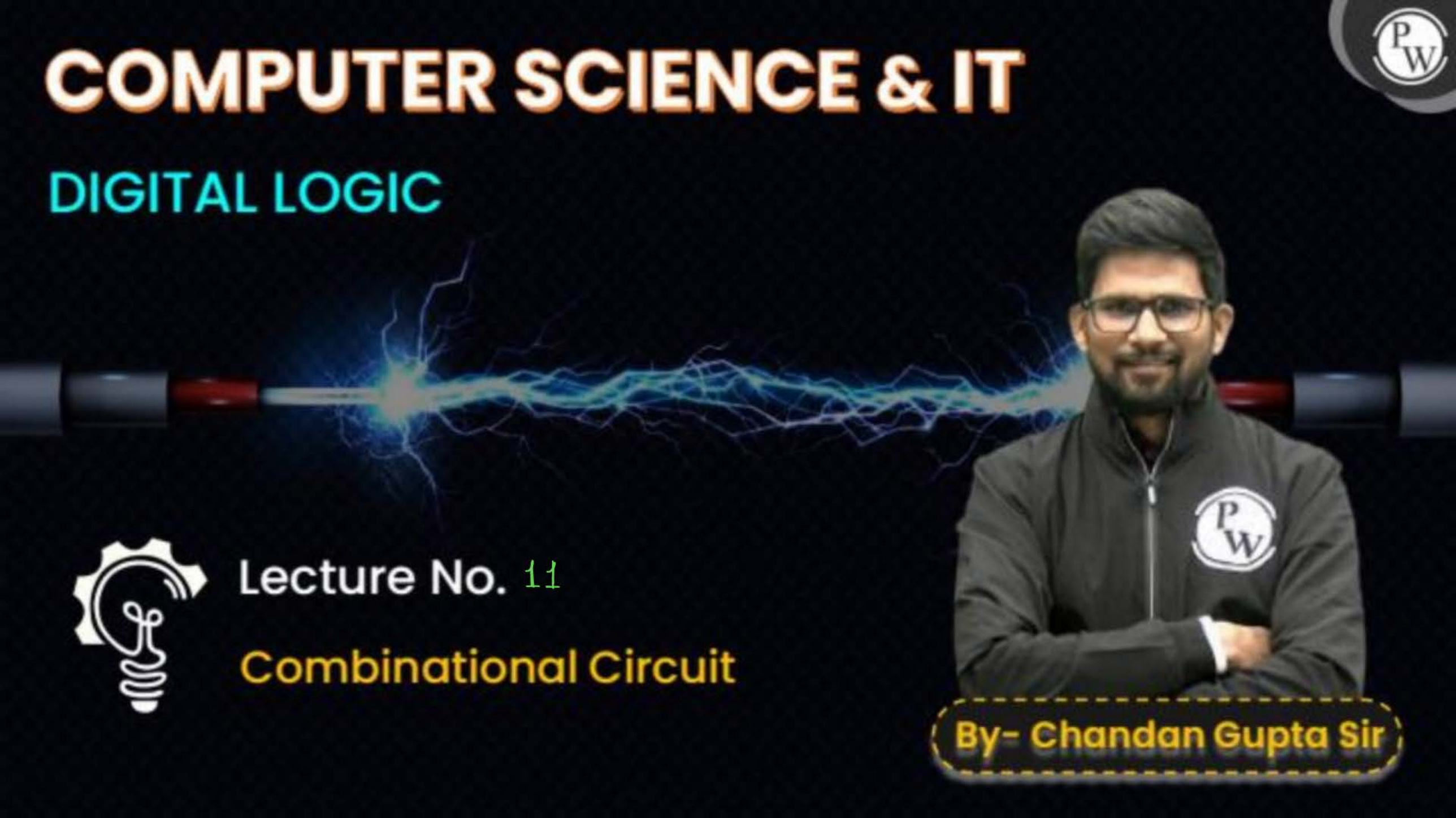
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 11

Combinational Circuit

By- Chandan Gupta Sir



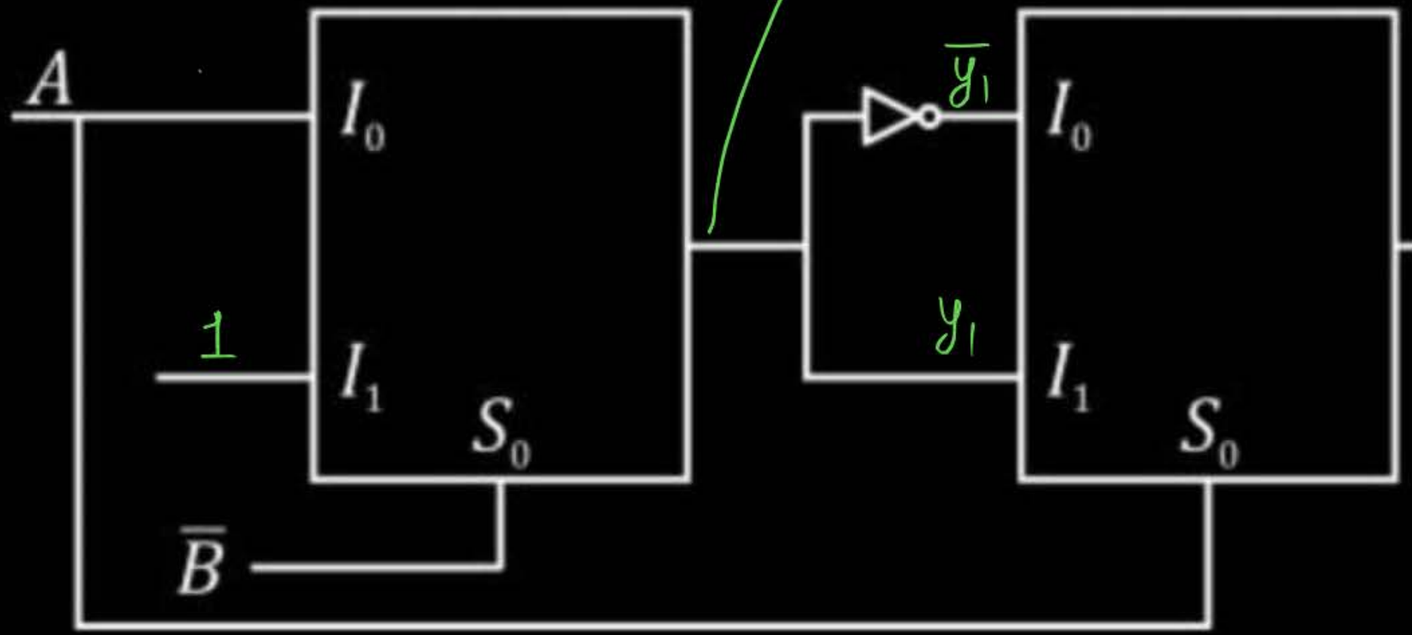
MUX +
Question Discussion



- Question Discussion on MUX.

Question 5

A digital circuit is as given below :



$y_1 = \bar{S_0} \cdot A + S_0 \cdot 1 = B \cdot A + \bar{B}$
$= (A + \bar{B})$

$y = y_1 \odot A$
$= (A + \bar{B}) \odot A$
$= \overline{(A + \bar{B})} \cdot \bar{A}$
$+ (A + \bar{B}) \cdot A$
$= \bar{A} \cdot B\bar{A} + A$
$= \bar{A} \cdot B + A$
$= (A + B)$

The output y is

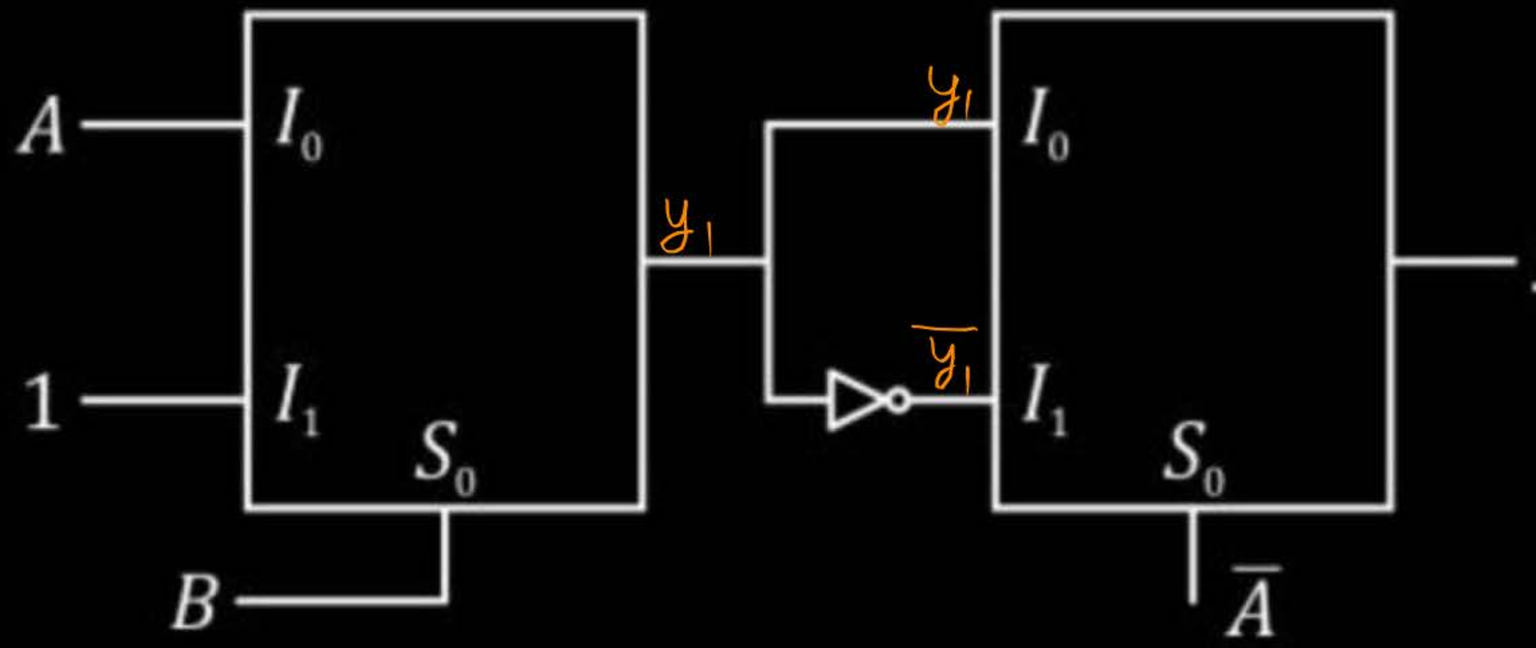
(a) $A \odot B$ (b) $\bar{A}B$

(c) $(A + B)$ (d) $A\bar{B}$

Question 6

$y_1 = (\overline{B} \cdot A) + B \cdot 1 = (A + B)$

A digital circuit is as given below :



$y = y_1 \oplus \overline{A}$

$= (A+B) \oplus \overline{A}$

$= \overline{(A+B)} \cdot \overline{A} + (A+B) A$

$= \overline{A} \cdot \overline{B} + A$

$= A + (\overline{A} \cdot \overline{B}) = (A + \overline{B})$

Output y is

(a) 1 ✗

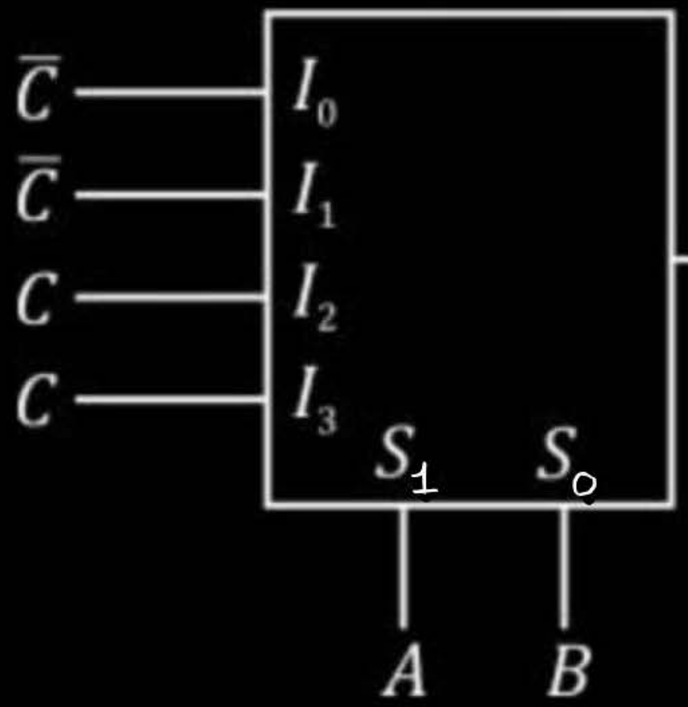
(b) ✗ $A \odot B$

(c) $\overline{\overline{A}B} = (A + \overline{B})$

(d) $\overline{A \cdot B}$

Question 7

A digital circuit is as given below :



$y =$

$A=0, B=0$

$A=0, B=1$

$\overline{A}\,\overline{C}\,[\overline{B}+B] = \overline{A}\,\overline{C}$

$\overline{A}\,\overline{B}\cdot\overline{C} + \overline{A}B\cdot\overline{C} + A\overline{B}\cdot C$

$+\overline{A}B\cdot C$

$AC(\overline{B}+B) = AC$

$= \overline{A}\,\overline{C} + AC$

$= A \odot C$

Output y is

(a) ✗ $A \oplus B \oplus C$ C

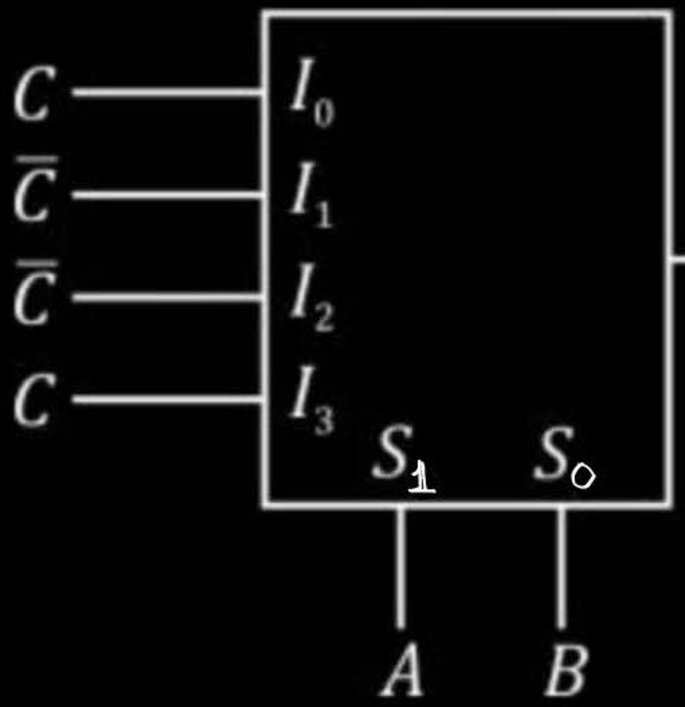
(b) ✓ $A \odot C$ $\overline{C}, \overline{C}$

(c) ✗ $A \odot B \oplus C$ $\overline{C}, C$

(d) ✗ $A \oplus B$ 0

Question 8

A digital circuit is as given below :



Output y is

(a) $A \oplus B \oplus C$

(b) $A \odot B \oplus C$

(c) $\overline{A \oplus B \odot C}$

(d) $A \oplus B$

$\bar{B}\bar{Q} + BQ$

$= B \odot Q$

$= B \odot A \odot C$

$A = 0, B = 0$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$		1		1
A	1		1	

$y = \bar{A}\bar{B}\cdot C + \bar{A}B\cdot\bar{C} + A\bar{B}\cdot\bar{C} + AB\cdot C$

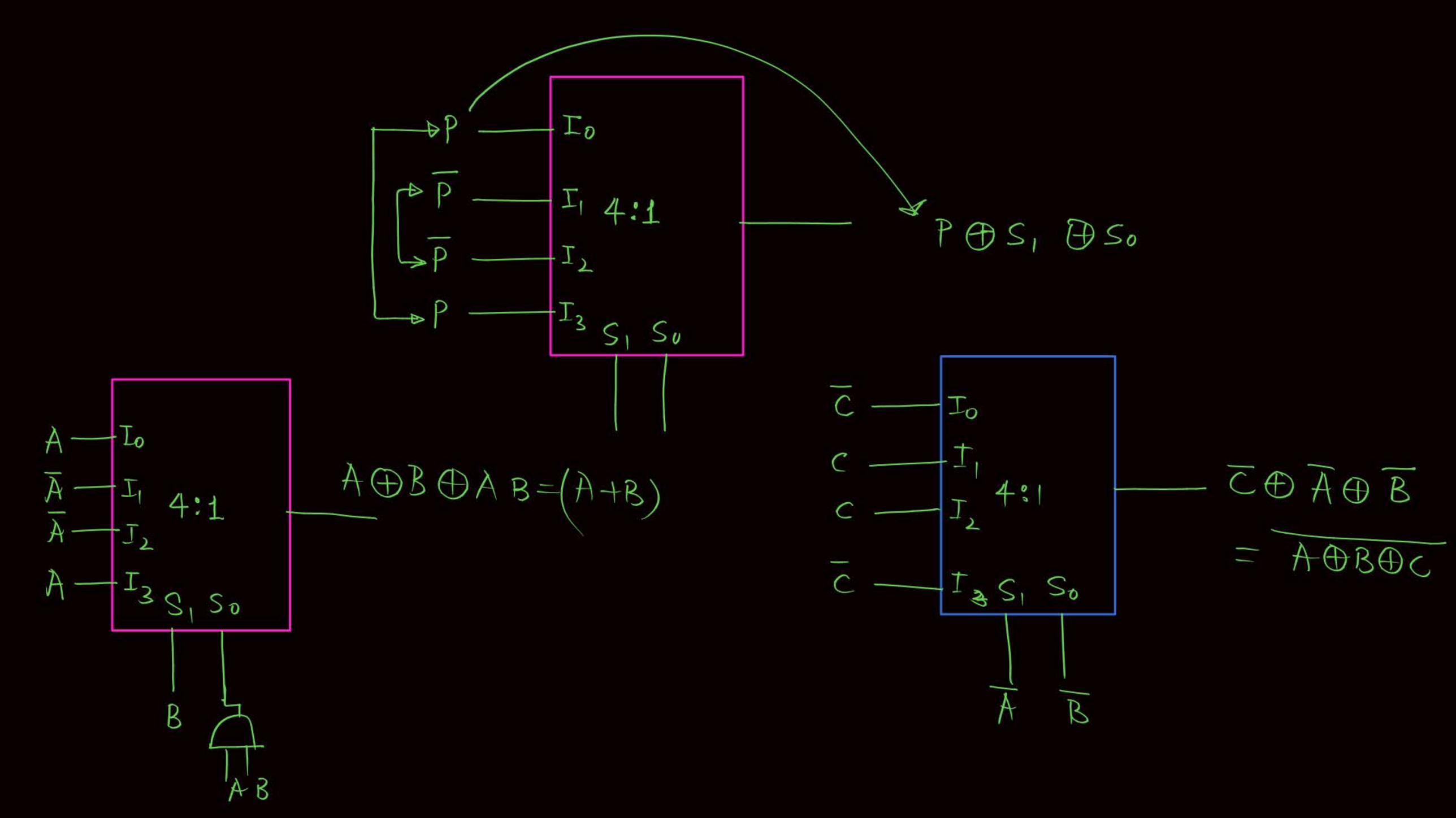
$= \sum(1,2,4,7) = A \oplus B \oplus C$

$= \bar{B}[\bar{A}C + A\bar{C}] + B[\bar{A}\bar{C} + AC]$

$= \bar{B}[A \oplus C] + B[A \odot C]$

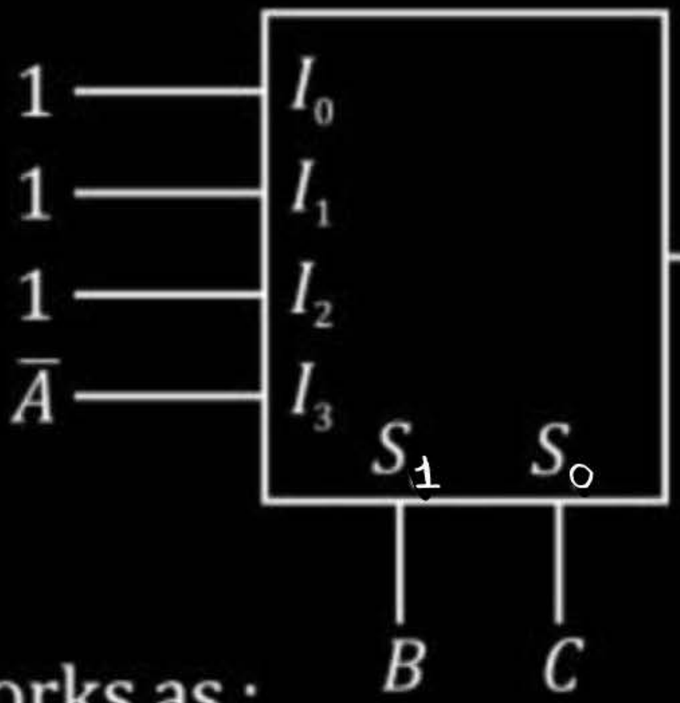
$= P\cdot\bar{B} + \bar{P}\cdot B = P \oplus B$

$= A \oplus C \oplus B$


P
$\bar{P}$
$\bar{P}$
P
I_0
I_1
I_2
I_3
4:1
S_1 S_0
$P \oplus S_1 \oplus S_0$
A
$\bar{A}$
$\bar{A}$
A
I_0
I_1
I_2
I_3
4:1
S_1 S_0
B
A B
$A \oplus B \oplus A\,B = (A + B)$
$\bar{C}$
C
C
$\bar{C}$
I_0
I_1
I_2
I_3
4:1
S_1 S_0
$\bar{A}$ $\bar{B}$
$\bar{C} \oplus \bar{A} \oplus \bar{B}$
$= \overline{A \oplus B \oplus C}$

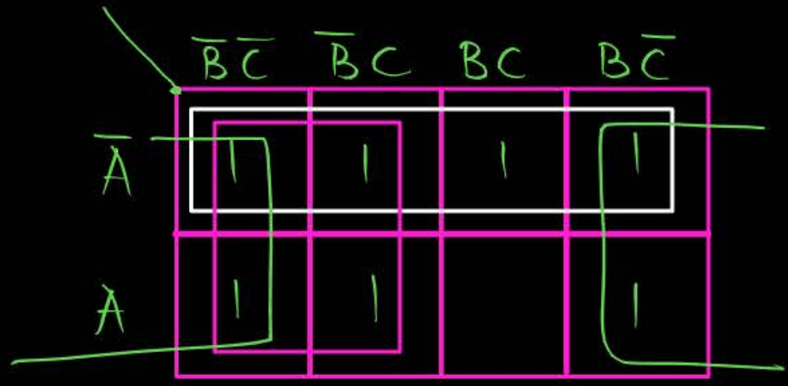
Question 9

A digital circuit is as given below :



B=0, C=0

$y = \bar{B}\bar{C}\cdot 1 + \bar{B}C\cdot 1 + B\bar{C}\cdot 1 + BC\cdot\bar{A}$



$= \pi(7)$

$= (\bar{A}+\bar{B}+\bar{C})$

$= \overline{A\cdot B\cdot C}$

Above circuit works as :

(a) ✗ 3-input OR gate $A+0+0 = A$ (b) ✗ 3-input NOR gate $\overline{A+0+0} = \bar{A}$

(c) ✗ 3-input AND gate $A\cdot 0\cdot 0 = 0$ (d) 3-input NAND gate $\overline{A\cdot 0\cdot 0} = 1$

Question 10

A digital circuit is as given below :

Output y will be : $Y[A, B, C, D]$

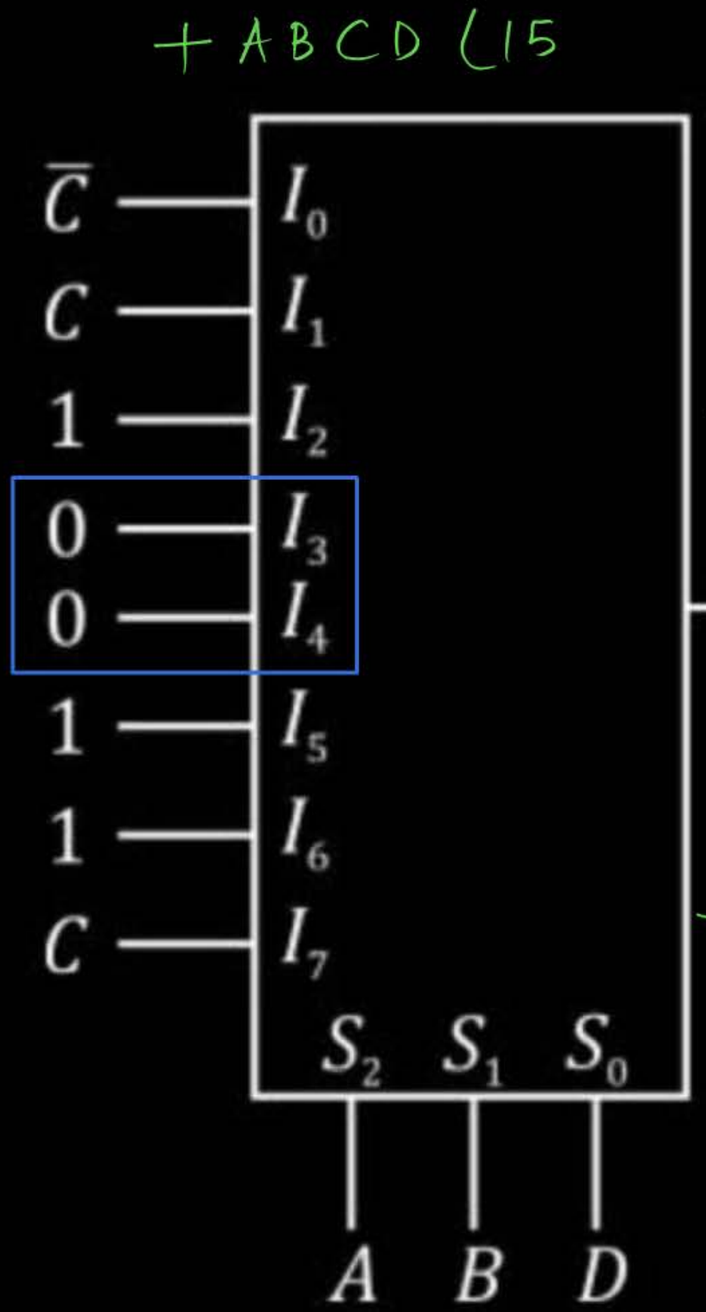
(a) $y = \Sigma\,(0,3,4,6,7,9,11,12,14,15)$

(b) $y = \Sigma\,(0,1,3,4,6,8,9,11,12,14,15)$

(c) $y = \Sigma\,(0,3,4,6,9,11,12,14,15)$

(d) $y = \Sigma\,(0,3,4,6,9,10,11,12,14,15)$

$\Sigma(0, 3, 4, 6, 9, 11, 12, 14, 15)$



$+ABCD$ (15

$+AB\bar{D}$ (12, 14)

$+\bar{A}BD \cdot 0$

$+A\bar{B}\bar{D} \cdot 0$

$+A\bar{B} \cdot 1D$ (9, 11)

$= \bar{A}\bar{B}\bar{C}\bar{D}$ (0)

$+\bar{A}\bar{B}CD$ (3)

$+\bar{A}B\bar{D}\,1$ (4, 6)

$\overline{A}\,B\,D$

$0\ 1\ 0\ 1 \rightarrow 5$

$0\ 1\ 1\ 1 \rightarrow 7$

$A\,\overline{B}\,\overline{D}$

$1\ 0\ 0\ 0 \rightarrow 8$

$1\ 0\ 1\ 0 \rightarrow 10$

Question 11

A 4 : 1 multiplexer is used for generating output carry of a full adder. A and B are bits to be added while C_{in} is input carry and C_{out} is output carry. Which of the following is choice of signals at I_0, I_1, I_2, I_3.

(a) $I_0 = 0, I_1 = C_{in}, I_2 = C_{in}, I_3 = 1$

(b) $I_0 = 1, I_1 = C_{in}, I_2 = C_{in}, I_3 = 1$

(c) $I_0 = C_{in}, I_1 = 0, I_2 = 1, I_3 = C_{in}$

(d) $I_0 = 0, I_1 = C_{in}, I_2 = 1, I_3 = C_{in}$

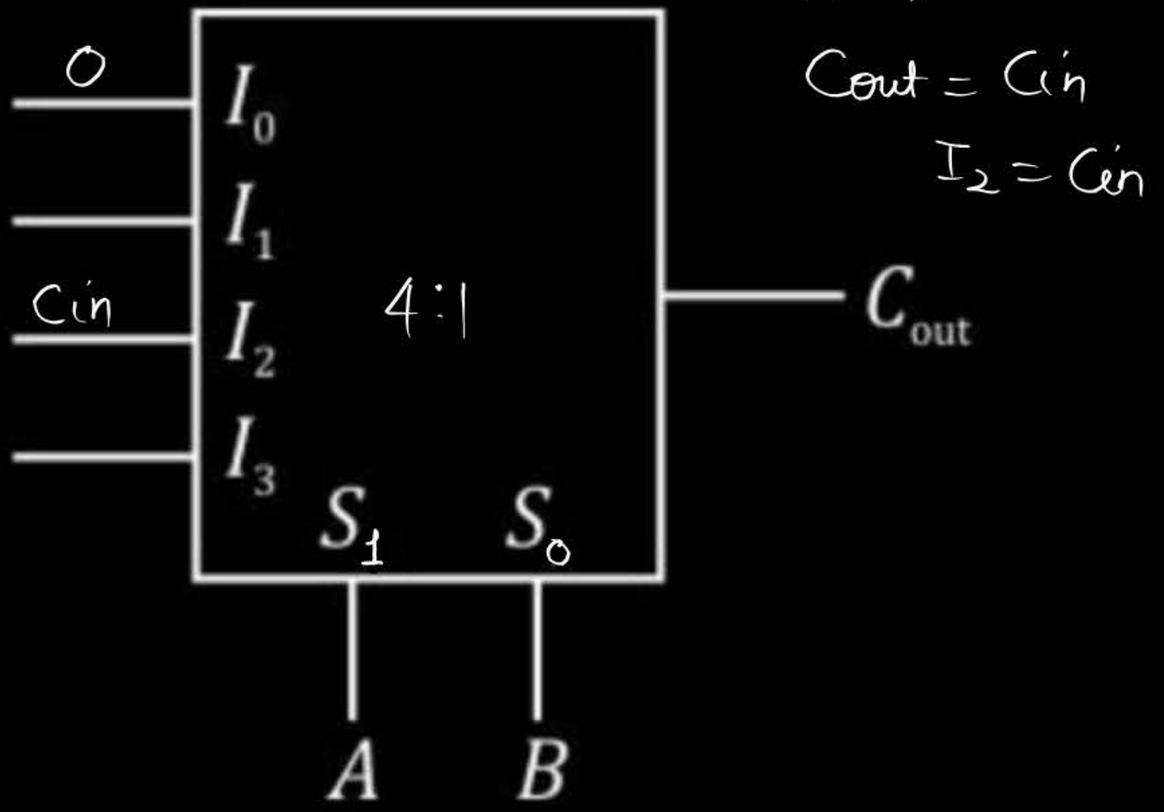
$C_{out} = AB + BC_{in} + AC_{in} = \Sigma(3,5,6,7)$

(A, B, C_{in})

A=1, B=0

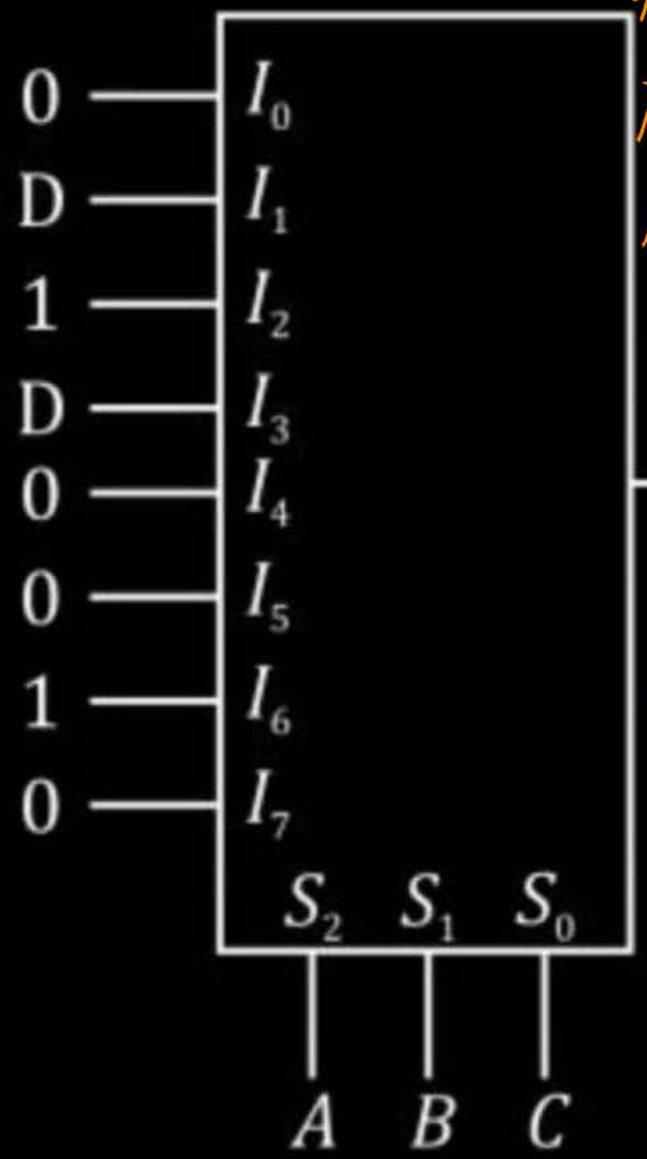
$C_{out} = C_{in}$

$I_2 = C_{in}$



Question 12

A digital circuit is as given below :



$y\,(A, B, C, D)$ will be

(a) $A\bar{B}C + A\bar{C}D$ — 0 , 0

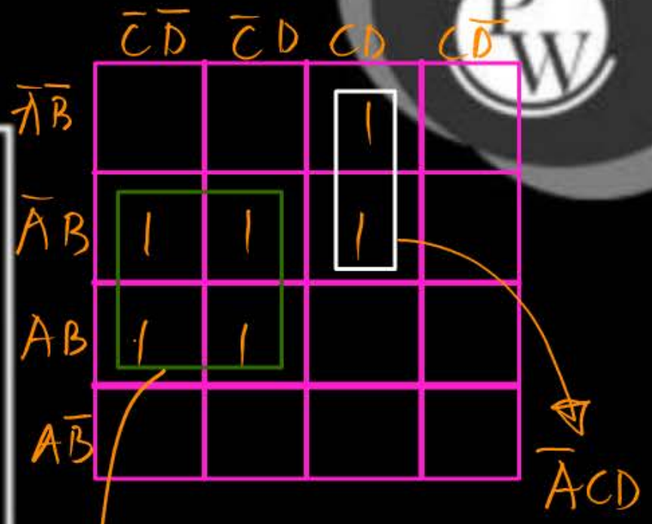
(b) $BC + A\bar{B}D$ — 0 , 1

(c) $B\bar{C} + \bar{A}CD$ — 0 , D

(d) $\bar{B}C + A\bar{B}D$ — 0 , 0

A=0, B=0, C=0

A=0, B=1, C=1



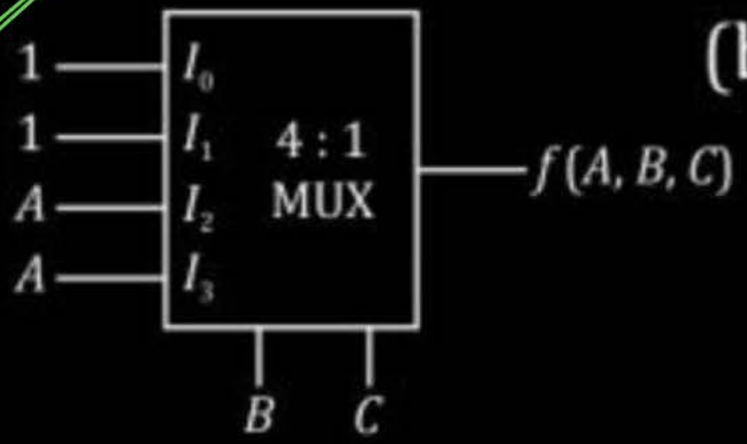
$y = \bar{A}\bar{B}C\cdot D + \bar{A}B\bar{C}\cdot 1 + \bar{A}BC\cdot D + AB\bar{C}\cdot 1$

Question 13

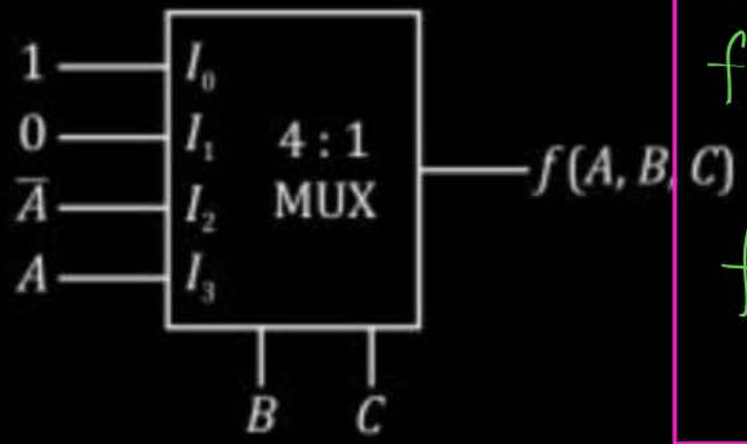
A logical function is given as : $f(A, B, C) = \overline{\bar{A}B + B\bar{C}} + AB$

Its implementation using MUX is given below. Then which implementation is correct

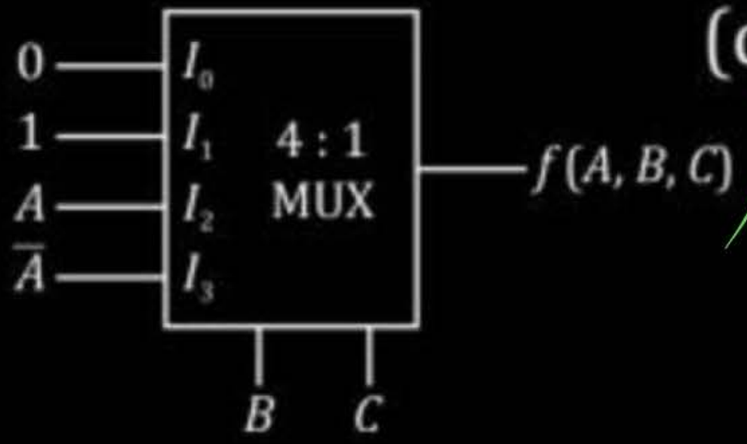
(a)



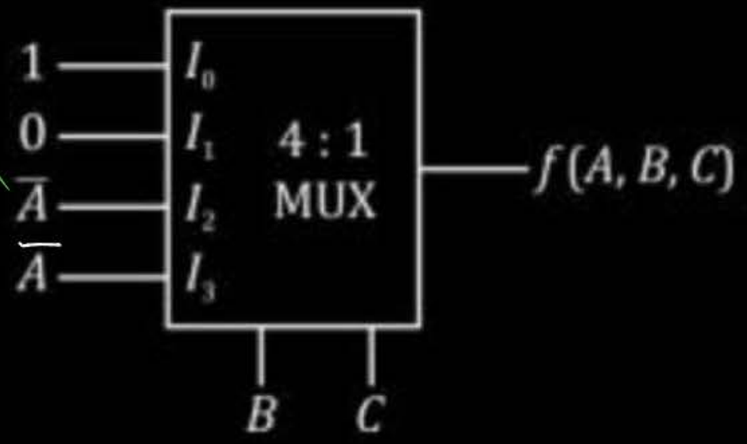
(b)



(c)



(d)

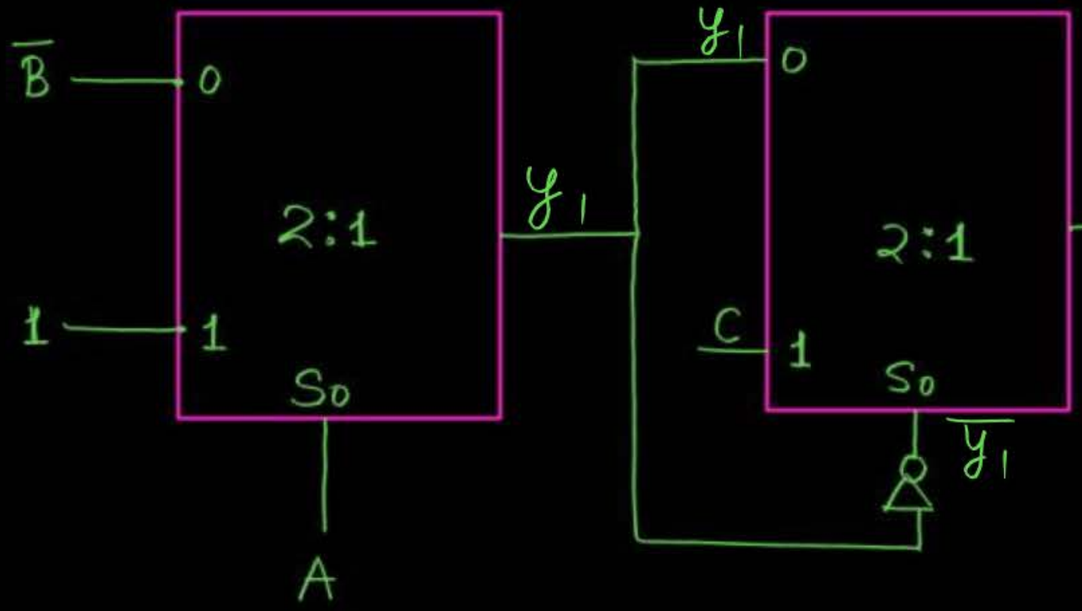


B=0, C=0

$f(A, 0, 0) = 1$, $I_0 = 1$

$f(A, 0, 1) = 1$, $I_1 = 1$

#Q. A MUX ckt is as given below:



$y_1 = \overline{A} \cdot \overline{B} + A \cdot 1 = (\overline{A} + \overline{B})$

Answer (7)

$y = \overline{S_0} \cdot y_1 + S_0 \cdot C$

$= y_1 \cdot y_1 + (\overline{y_1} \cdot C)$

$= (y_1 + \overline{y_1}) \cdot (y_1 + C) = y_1 + C$

$= (A + \overline{B} + C)$

0 1 0

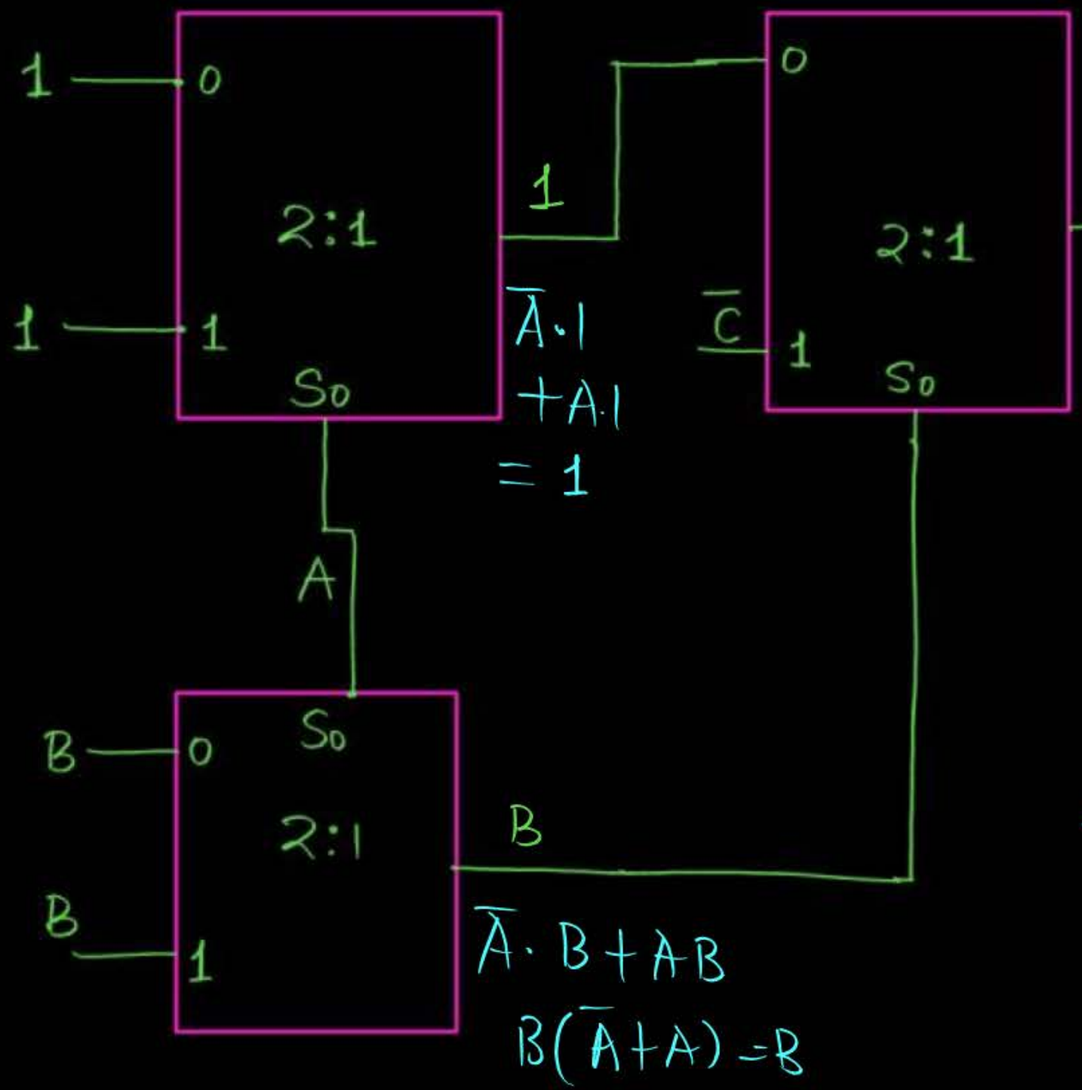
The no. of distinct combination of A, B & C for which o/p y will be 1 is ____.

$= \pi(2)$

$\Sigma(0,1,3,4,5,6,7)$

A		$\overline{B}$		C	
1	00 → 4	0 0 0	→ 0	0 0 1	→ 1
1	01 → 5	0 0 1	→ 1	0 1 1	→ 3
1	10 → 6	1 0 0	→ 4	1 0 1	→ 5
1	11 → 7	1 0 1	→ 5	1 1 1	→ 7

#Q. A MUX CKt is as given below:



No. of different combination of A, B, C for which O/P y will be 1 ____6____.

$y = \bar{B}\cdot 1 + B\cdot\bar{C} = (\bar{B}+\bar{C})$

$\bar{B}$

0 0 0 → 0
0 0 1 → 1
1 0 0 → 4
1 0 1 → 5

$B\bar{C}$

0 1 0 → 2
1 1 0 → 6

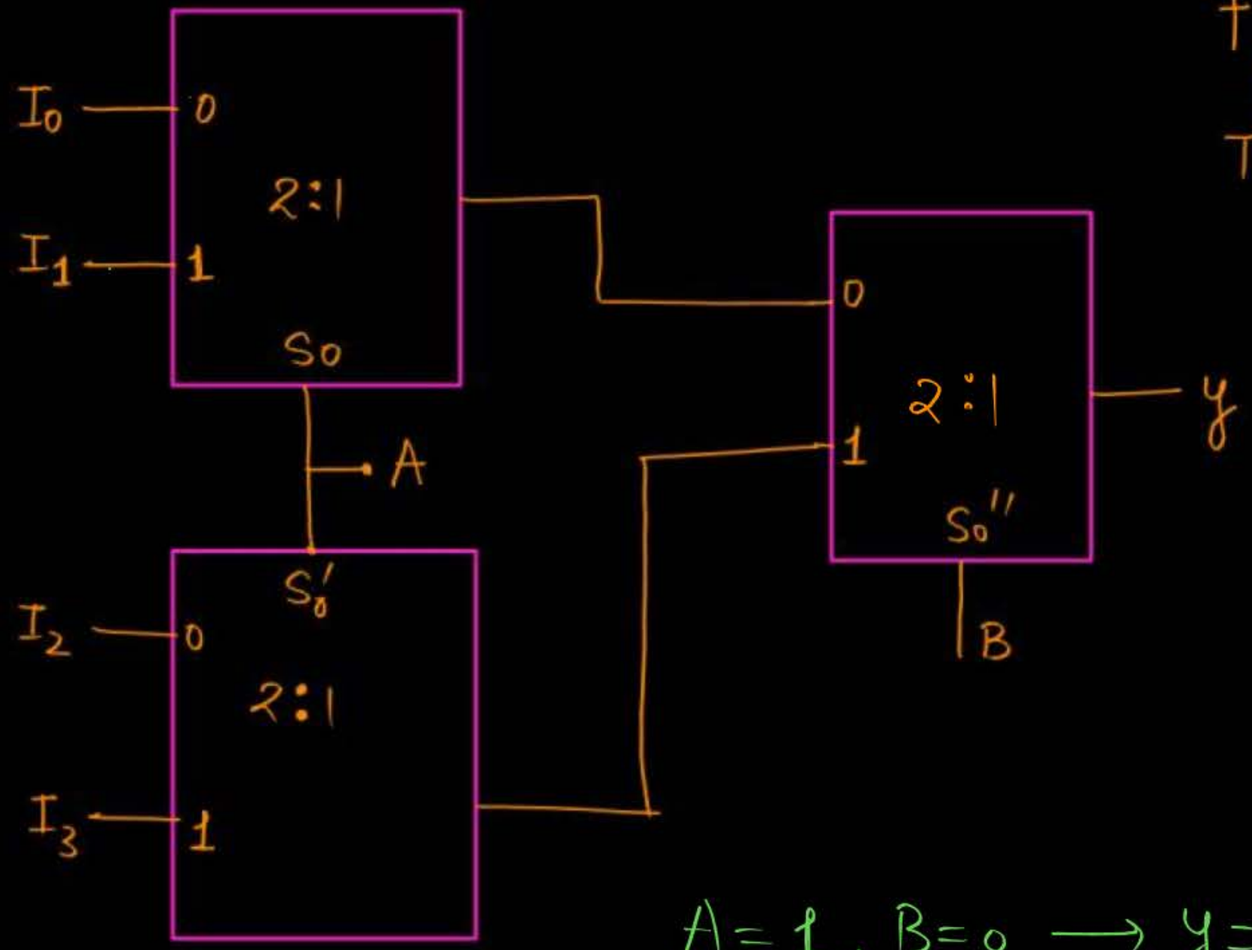
$= \Sigma(0,1,2,4,5,6)$

$(\bar{B}+\bar{C})$

0 1 1 → 3
1 1 1 → 7

→ POS.

#Q. A MUX CKt is as given below:



We want to implement a function $f(A,B,C) = \Sigma(0,1,4,6,7)$ using given MUX CKt.

Then i/p lines I_1 & I_2 must be

a. $I_1 = 0$, $I_2 = \overline{C}$

b. $I_1 = 0$, $I_2 = C$

c. $I_1 = \overline{C}$, $I_2 = C$

d. $I_1 = \overline{C}$, $I_2 = 0$

$A=1, B=0 \longrightarrow y = I_1$, $f(1,0,C) = \Sigma 4 = A\overline{B}\,\overline{C} = \overline{C}$

$I_1 = \overline{C}$

$A=0, B=1 \longrightarrow f(0,1,C) = 0$, $I_2 = 0$

[MSQ] [1 Mark]

#Q. Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a,b,c,d,e are Boolean variables?

$X(a,b,c) = ab + bc + ca$

$= a\cdot b + b\cdot[ab+bc+ca] + a[ab+bc+ca] = ab + ac$

$ab + ab + bc + ab + ac = ab + bc + ac$

(A) $X(a,b,X(c,d,e)) = X(X(a,b,c),d,e)$

$a=1, b=1$ $X(c,d,e) = X$,

$X[a,b,X(c,d,e)] = 1$

$a=1, b=1, c=0, d=0, e=0$

$1, \quad X[1,0,0] = 0$

(B) $X(a,b,X(a,b,c)) = X(a,b,c)$

(C) $X(a,b,X(a,c,d)) = (X(a,b,a) \text{ AND } X(c,d,c))$

0 1 0 1 1 0

$1 \quad ab + bc + abc = ab + bc$

(D) $X(a,b,c) = X(a,X(a,b,c),X(a,c,c))$

© $X[a,b,a] = a$

$X[c,d,c] = c$

$X[a,b,a]$ AND $X[c,d,c] = a \cdot c$

$a \cdot b + b \cdot [ac + cd + ad] + a[ac + cd + ad]$

$X[a, X(a,b,c), X(a,c,c)] =$

$X[a, X(a,b,c), c]$

$= a[ab + bc + ac] + c[ab + bc + ac] + ac$

$= ab + ac + bc + ac + ac$

$= ab + bc + ac$

H.A. and F.A. using 2 : 1 MUX

- H.A. using 2 : 1 MUX :

H.W.

- F.A. using 2 : 1 MUX :

H.W.



2 Minute Summary

→ Question Discussion.

Thank you

GW

Soldiers !

